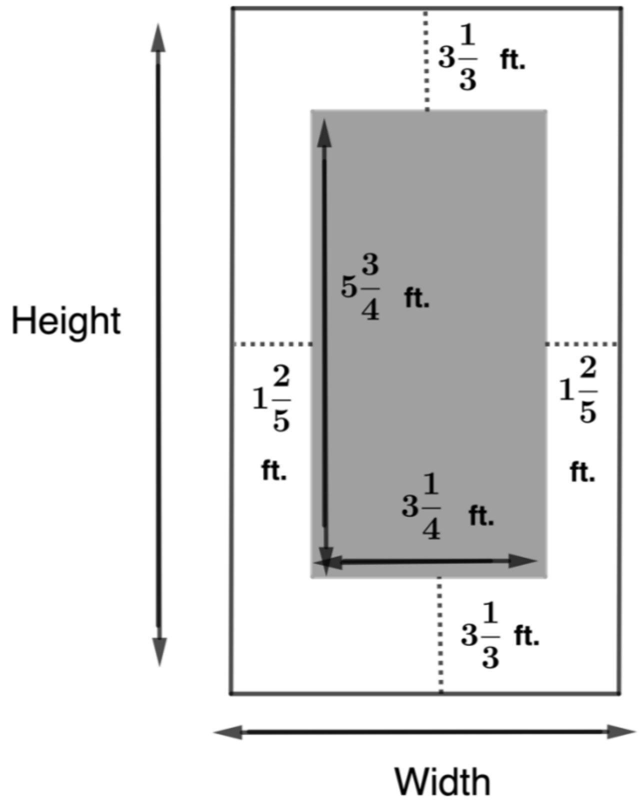
Grade 6
Unit 1
Lesson 2 Practice

Name___Answer Key______
Date____________________
Period__________________

1. What is the common denominator of $4\frac{1}{7}$ and $2\frac{3}{8}$? 56

Use the given context and image to answer questions 2-7.

Sawyer's mom gifted her with a framed quilt for her birthday. The dimensions are shown in the image, in feet (ft).



2. What is the perimeter of the quilt, in ft?

$P = l + l + w + w$ or $P = 2l + 2w$
$P = 5\frac{3}{4} + 5\frac{3}{4} + 3\frac{1}{4} + 3\frac{1}{4}$
$= 16\frac{8}{4}$
$= 16 + 2 = 18$ ft

3. Write an expression that represents the width of the frame and quilt.

$1\frac{2}{5} + 3\frac{1}{4} + 1\frac{2}{5}$

4. What is the total width, in ft?

$W = 1\frac{2}{5} + 3\frac{1}{4} + 1\frac{2}{5}$
$= 1\frac{8}{20} + 3\frac{5}{20} + 1\frac{8}{20}$
$= 5\frac{21}{20} = 6\frac{1}{20}$ ft

5. Write an expression that represents the height of the frame and quilt.

$3\frac{1}{3} + 5\frac{3}{4} + 3\frac{1}{3}$

6. What is the total height, in ft?

$$h = 3\frac{1}{3} + 5\frac{3}{4} + 3\frac{1}{3}$$
$$= 3\frac{4}{12} + 5\frac{9}{12} + 3\frac{4}{12}$$
$$= 11\frac{17}{12}$$
$$h = 12\tfrac{5}{12} \text{ ft}$$

7. What is the total perimeter of the framed quilt, in ft?

$$P = 2h + 2w$$
$$= 2\left(12\frac{5}{12}\right) + 2\left(6\frac{1}{20}\right)$$
$$= 24\frac{10}{12} + 12\frac{2}{20}$$
$$= 24\tfrac{50}{60} + 12\tfrac{6}{60} = 36\tfrac{56}{60} = 36\tfrac{14}{15} \text{ ft}$$

For questions 8-10, rewrite the expression as an expression with common denominators. Then, add or subtract the values.

8. $6\frac{1}{8} - 2\frac{3}{5} = \frac{49}{8} - \frac{13}{5} = \frac{245}{40} - \frac{104}{40} = \frac{141}{40} = 3\frac{21}{40}$

9. $4\frac{1}{4} + 6\frac{7}{10} = \frac{17}{4} + \frac{67}{10} = \frac{85}{20} + \frac{134}{20} = \frac{219}{20} = 10\frac{19}{20}$

10. $2\frac{1}{9} - \frac{4}{7} = \frac{19}{9} - \frac{4}{7} = \frac{133}{63} - \frac{36}{63} = \frac{97}{63} = 1\frac{34}{63}$